Pakistan Journal of Pharmacol. ISSN 0255-7088
Vol. 15(1), January 1998, pp.55-59

NUCLEIC ACID CONTENTS IN CATARACTOUS PATIENTS

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ABSTRACT:

The study incudes 35 patients suffering from cataract and 17 control subjects. DNA in the blood of cataractous patients is significantly low as compared to control subjects whereas RNA content is significantly high in patients as compared to control subjects. The lenticular tissue has shown no significant difference in the content of DNA in different types of cataract.

INTRODUCTION

Cataract is one of the leading cause of blindness (Taylor *et al.*, 1991). Opacities of the lens i.e. cataract, is a disease of aging humans, which deprives them of visual activity (Hoenders and Bloemendel 1981). It is generally accepted that oxidation of lens protein is an important event in the formation of cataract (Truscott and Augesteyn 1977). However, there are many other factors that may cause cataract such as genetic and metabolic disorders, intraocular diseases, dietary deficiency and old age (Hoenders and Bloemendel 1981).

The ageing of human lens is characterized by an increase in insoluble proteins i.e. aggregation of high molecular weight lens proteins and a decrease in low molecular weight proteins (Augusteyn 1981). The conversion of soluble protein to insoluble form is associated with yellow coloration and non-tryptophan fluorescence which may lead to nuclear cataract.

Bloemendel (1984) suggested that in cataracts, the first hit is always at the level of DNA, and the changes to proteins and metabolism are only secondary effects. Mitomycin C, a DNA synthesis inhibitor, induced cataract in rat lenses in vitro (Mikuni 1980). DNA damage, repair and replication in selinite induced cataract in rat lens was evaluated by Huang *et al.* (1990). Each component of DNA was affected by the occurrence of selenite stress in lens. The morphological changes suggest an underlying damage to DNA (Frankel and Falvey 1988). The cataract or lens opacity mutations is an amino-acid substitution that inhibits targeting of major intrinsic proteins to the cell membrane. These allelic cataract mutations provide the direct evidence that major intrinsic proteins play a crucial role in the development of transparent eye lens (Shiels and Bassnett 1996). These results support a role of abnormal DNA metabolism in cataractogenesis, most probably a factor in the later stage of cortical cataract (Huang *et al.*, 1990).

Specific genes and their transcripts have been detected in various tissues and cells by in situ hybridization with labelled RNA probes (Cox *et al.*, 1984). MP26 messenger RNA constitute 1 to 1.5% of the total mRNA population (Ming-Youn *et al.*, 1987). Bekhor (1988) have shown that aldose reductase in mRNA is increased in rat lens epithelial cells in developing cataracts.

MATERIALS AND METHODS

Human cataractous lenses were collected at the time of operation from the patients of the Eye Department of Abbasi Shaheed Hospital, Karachi. The patients were examined to determine the type of cataract. The fasting blood sample of the same patients (age ranges from 36 to 60 years) were collected in the heparinized tubes. After lentectomy the lenses were placed in a container filled with chilled Tris buffer (pH 7.8) and then they were kept at -20^{o}C in the deep freezer till analysis.

Each lens was dried and then weighted and was then homogenized. 5% homogenate was prepared in cold Tris buffer (pH 7.8). The aliquotes of lens and the blood were used for the estimation of glucose (Dubowski 1962), protein (Reinhold 1963 cited by Varley 1969), RNA (Thanhauser 1945) and DNA (Burton 1968).

The blood of control subject (age ranges from 21 to 60 years) having no ophthalmic disease were also collected and was analyzed for glucose, protein, RNA and DNA.

RESULTS

In this study 35 cataractous patients (20 male and 5 female) with different type of cataracts has been investigated. The duration of disease ranged from months to years. Seventeen control (7 male and 10 female) subjects having no ophthalmic or any disease and had no family history of cataract were also included in the study. Table-1 shows the blood level of DNA which is significantly low ($P<0.001$) and RNA level significantly high ($P<0.001$) in cataractous patients as compared to control subjects.

The cataract lenses have been classified on the basis of opacification, partially opacified as immature, completely opacified as mature and lenses showing degenerative changes in fiber with complete opacification as hypermature. The patients suffering from diabetes as diabetics.

Table-2 shows the level of lenticular glucose is significantly ($P<0.001$) increased in diabetic lenses as compared to the mature lenses. Lens total protein is significantly decreased from mature to diabetic lens ($P<0.001$) when compared with mature cataract lens, whereas no significant change is observed in the mean values of DNA.

DISCUSSION

Cataract or age-related eye lens opacification is one of the most common disease of eye and the major cause of blindness worldwide.

Sugar cataracts were presumed to result from an increase in the oxidation potential of the lens fiber cells leading to fiber cell destruction and crystalline and non-crystalline protein aggregation (Unakar *et al.*, 1978). In the present study an increased level of lens glucose has been found in diabetic cataractous lens as compared to mature lens. Similar results has been reported by Hussain (1989), Siddiqui (1991) and Zafar (1992). This may be because glucose utilization is slow in diabetic than in normal (Forbath and Heleny 1968). The glucose and fructose binds with serum albumin in a non-enzymatic glycation reaction demonstrate the ability of these sugars to modify proteins in vitro (Suarez *et al.*, 1989).

A significant decrease in the protein concentration in the diabetic lens as compared to the mature cataractous lens has been found as was found by Sarwat (1985) and Hussain (1989), who had shown that the concentration of protein decreases with the progression of cataractogenesis. The decreased protein content of the crystalline lens may be due to the leakage of low molecular weight proteins from lens membrane to the aqueous homour (Fujiwara 1989). Kuharana *et al.* (1982) also found that total protein and amino acid content of the crystalline lens decreased with the maturation of cortical cataract with an increase content of protein in the aqueous humour. Rawal and Gandhi (1986) had shown a significant decrease of DNA in blood and lens at different periods of alloxan-treatment. Shearer *et al.* (1987) have suggested that the cortical opacity is due to disturbed fibrogenesis. These morphological changes result in an underlying damage to DNA. In the present study a significantly decreased level of DNA has been found in the blood of cataractous patients as compared to the normal subjects, confirming the results shown by Rawal and Gandhi (1986). In the present study a significant increase in the RNA concentration in mature cataractous lens has been found as compared to diabetic lens. If synthesis of RNA in mature cataracts is enhanced as previously suggested (Ming-Youn *et al.*, 1987), then it may be postulated that at least in the case of galactose cataracts the cortical fiber cells may be undergoing a process of attempted recovery. This is in fact the case when galactose-cataract is reversed (Bekhor 1988). In the present study a significantly increased level of RNA has been found in the blood of cataractous patients as compared to the normal control subjects.

Table 1
Variation of Blood Constituents in the Control and Cataractous Patients

Type	Glucose (mg/dl)	Total protein (g/dl)	Albumin (g/dl)	A/G ratio	DNA (μg/dl)	RNA (μg/dl)
Control	104.70	7.35	4.27	1.42	36.41	38.10
	± 1.04	± 0.19	± 0.17	± 0.09	± 0.55	± 0.93
	(17)	(17)	(17)	(17)	(17)	(17)
Patient	109.19	7.78	4.68	1.49	15.08	72.74
	± 1.41	± 1.15	± 0.13	± 0.07	± 0.37*	± 1.03*
	(35)	(35)	(35)	(35)	(35)	(35)

*$P < 0.001$ statistically significant values of the patient as compared to the control.

Table 2
Variation of Lenticular Constituents in the Different Type of Cataracts

Type	Glucose (mg/100 g)	Total protein (g/100 g)	Albumin (g/100 g)	A/G ratio	DNA (μg/100 g)	RNA (μg/100 g)
Mature	105.47	51.83	22.52	1.47	78.56	260.60
	± 1.26	± 1.21	± 0.48	± 0.09	± 1.16	± 2.56
	(28)	(28)	(28)	(28)	(28)	(28)
Hypermature	125.00	38.18	20.95	1.21	52.23	302.08
	± 0.00	± 0.69	± 0.15	± 0.18	± 1.86	± 9.90
	(2)	(2)	(2)	(2)	(2)	(2)
Diabetic	189.15	33.75	19.70	1.42	74.80	219.50
	± 3.48*	± 0.82**	± 0.67	± 0.09	± 2.60	± 5.23***
	(5)	(5)	(5)	(5)	(5)	(5)

* $P < 0.001$ statistically significant values of diabetic cataract lens as compared the mature cataract lens.

** $P < 0.001$ statistically significant values of diabetic cataract lens as compared mature cataract lens.

*** $P < 0.001$ statistically significant values of diabetic cataract lens as compared mature cataract lens.

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